AutiWise - A comprehensive digital toolkit for individuals with Autism Spectrum Disorder

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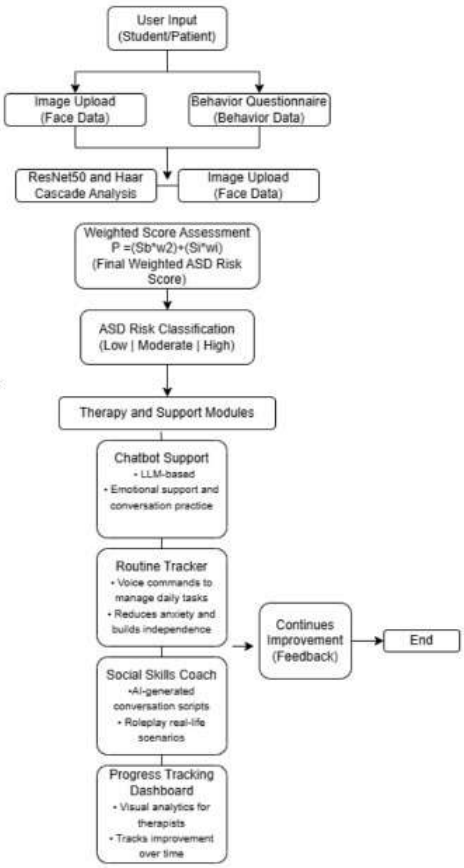
PROBLEM STATEMENT

Individuals with Autism Spectrum Disorder (ASD) face significant challenges in early diagnosis, effective communication, and social interaction. Current digital tools are often fragmented, addressing only one aspect of support. This leads to an inconsistent user experience and fragmented care. AutiWise seeks to solve this by developing a single, AI-powered digital toolkit that provides a holistic solution for individuals with ASD and their caregivers.

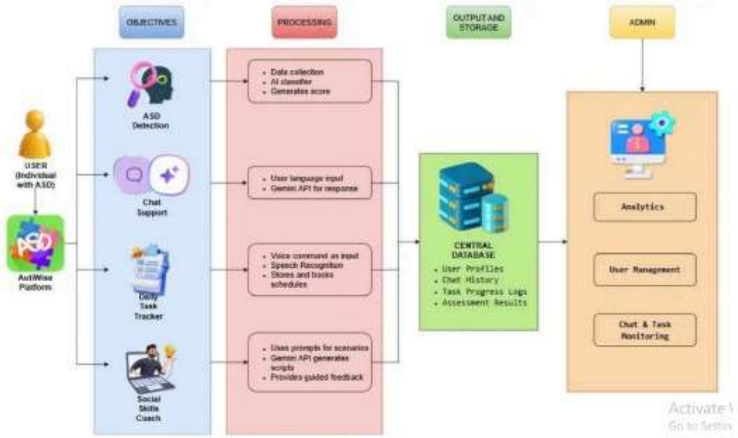
OBJECTIVES

- To design and implement an AI-powered system for the early detection of Autism Spectrum Disorder (ASD).
- To develop a conversational chatbot that facilitates communication support and emotional expression for individuals with ASD.
- To build a voice-command-based routine tracker that helps users create and follow structured daily schedules to reduce anxiety and improve time management.
- To create a social skills coaching module to generate real-world conversation scripts for role-play and social learning.

WORKFLOW



ARCHITECTURAL DIAGRAM



RESULT

A.Model Comparison and Accuracy Analysis

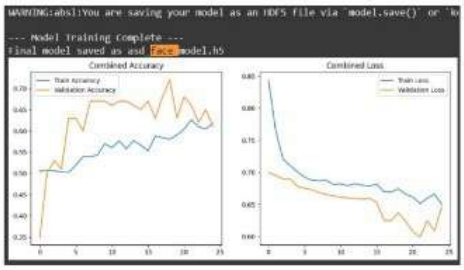


Fig1.ResNet50 Model Performance

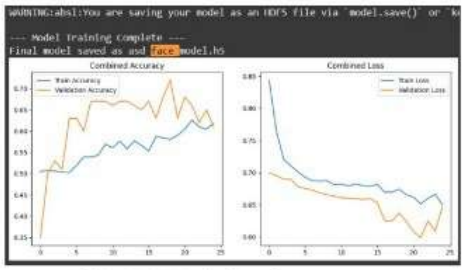


Fig2.CNN Model Performance

B.Multi-Mode ASD Synthesis Detection

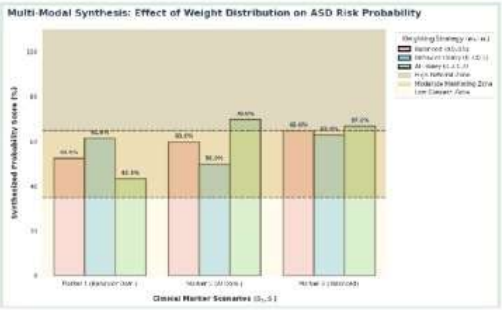


Fig3.SensitivityanalysisofASDriskprobabilityacrossvaryingweightageratios

C.NLP-Based Evaluation:Social Interaction

Example Phrase	Estimated Polarity	Estimated Subjectivity	System Feedback
"Is this seat taken?"	0.0 (Neutral)	0.0 (Factual)	Coaching Tip: This is a very practical and safe way to start.
"I really love your shirt!"	0.5 (Positive)	0.6 (Opinion)	Tone Check: Positive and friendly. Safe to compliment.
"This lunch is gross."	-0.7 (Negative)	0.8 (Opinion)	Tone: Negative. Might be perceived as forward or rude in a group.
"In for some conversation?"	0.0 (Neutral)	0.4 (Opinion)	Tone: Direct, Good. Straight to the point but may be forward.

Fig4. Sample sentiment classification and feedback and log in the social coach

CONCLUSION

- Within the framework of this project, an AI-powered integrated assistance system called AutiWise was presented and aimed at helping people with Autism Spectrum Disorder (ASD) and assisting therapists and caregivers in the process of continuous intervention. AutiWise provides the solutions to the key gaps existing in the current ASD care, including delayed diagnoses, insufficient communication support, and the absence of an organization of routine management.

REFERENCES

[1] Koegel, L.K., Ponder, E., Bruzzese, T. et al. Using Artificial Intelligence to Improve Empathetic Statements in Autistic Adolescents and Adults: A Randomized Clinical Trial. J Autism Dev Disord, (2025).

[2] Altın, Y., Bo¸snak, ¨ O. and Turhan, C. Examining Virtual Reality Interventions for Social Skills in Children with Autism Spectrum Disorder: A Systematic Review. J Autism Dev Disord, (2025).